Training Vectors: SVM QP Formulation

Training Vectors

$x_1 = (1, 1), \quad y_1 = 1$

$x_2 = (2, 2), \quad y_2 = 1$

$x_3 = (1, 4), \quad y_3 = -1$

$x_4 = (0, 4), \quad y_4 = -1$

Test Vectors

$3 \quad 3 \quad 2$

Use C=1

$\alpha_1 y_1 + \alpha_2 y_2 + \alpha_3 y_3 + \alpha_4 y_4 = 0$

$0 \le \alpha_1 \le 1$

$0 \le \alpha_2 \le 1$

$0 \le \alpha_3 \le 1$

$0 \le \alpha_4 \le 1,$

$$\text{Maximize } \mathcal{L}_p(\alpha) = \sum_{i=1} \alpha_i - \frac{1}{2} \sum_{i=1} \sum_{j=1} \alpha_i \alpha_j y_i y_j x_i^T x_j \Rightarrow \text{Minimize } \mathcal{L}_D(\alpha) = \frac{1}{2} \sum_{i=1} \sum_{j=1} \alpha_i \alpha_j y_i y_j x_i^T x_j - \sum_{i=1} \alpha_i$$

X $\quad$ $G = X \quad X$ $\quad$ $I = (y_i y_j)$ $\quad$ $I \cdot G$

$\lfloor 1 \quad 2 \quad 4 \quad 4 \rfloor$ $\quad$ $\lfloor x_4 \quad 1z \quad z_4 \quad \delta 0 \rfloor$ $\quad$ $\lfloor y_4 \quad -1 \quad -1 \quad 1 \quad 1 \rfloor$ $\quad$ $\lfloor y_4 \quad -1z \quad -z_4 \quad 1z \quad \delta 0 \rfloor$

From Previous Lecture

Minimize:

$$\lfloor +\alpha_1 + 2\alpha_1\alpha_2 - 3.5\alpha_1\alpha_3 - 6\alpha_1\alpha_4 + 2\alpha_2\alpha_1 + 4\alpha_2 - 11\alpha_2\alpha_3 - 12\alpha_2\alpha_4$$

$$-5.5\alpha_3\alpha_1 - 11\alpha_3\alpha_2 + 32.5\tilde{\alpha_3} + 36\alpha_3\alpha_4 - 6\alpha_4\alpha_1 - 12\alpha_4\alpha_2 + 36\alpha_4\alpha_3 + 40\tilde{\alpha_4}]$$

$$-\lfloor \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 \rfloor$$

$0 \le \alpha_1 \le 1$

$0 \le \alpha_2 \le 1$

$0 \le \alpha_3 \le 1$

$0 \le \alpha_4 \le 1$

Subject to:

$\sum_i \alpha_i y_i = 0 \qquad \alpha_1 + \alpha_2 - \alpha_3 - \alpha_4 = 0$

$\alpha_i \ge 0: \ \alpha_1 \ge 1, \ \alpha_2 \ge 1, \ \alpha_3 \ge 1, \ \alpha_4 \ge 1 \qquad \alpha_i \le 0: \ \alpha_1 \le 0, \ \alpha_2 \le 0, \ \alpha_3 \le 0, \ \alpha_4 \le 0$

Quadratic Equation in α with constraints